

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

What happens when an ASP.NET page is requested?



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can say what the ASP.NET life cycle specifies.
- I can talk about the stages in the application life cycle in ASP.NET.
- I can give precise information about the ASP.NET page life cycle at a work context.

Ask your partner:

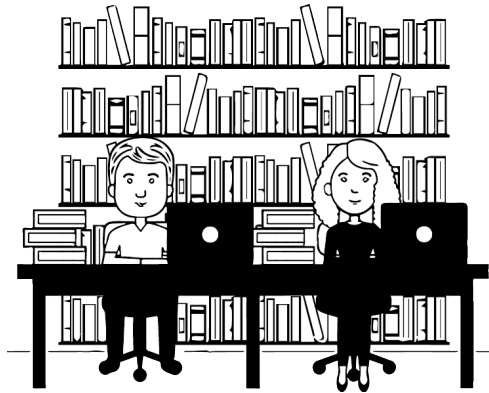
- What do you think are the processes involved when loading a web page?
- Do you believe that running a web page or an application have the same processing stages? Why?





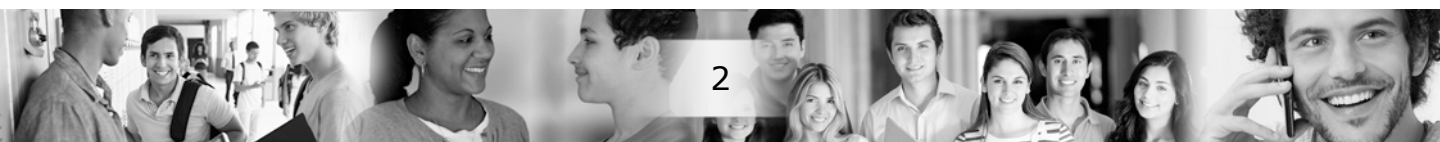
Let's Explore!

Jessica needs help with her homework!



Instructions: Read the text below.

- Jessica:** Hey, could you help me with my homework? I have to prepare a presentation about the ASP.NET life cycle.
- Nestor:** Sure, that's my favorite platform to assemble applications. What do you want to know?
- Jessica:** First, I need to say what the ASP.NET life cycle specifies.
- Nestor:** That's easy. There are three main aspects that are embedded; first, ASP.NET life cycle specifies the processes on the ASP.NET pages that produce dynamic output. Second, it tells us how applications and pages are processed and instantiated. Finally, it tracks pages and put them together in a dynamic way.
- Jessica:** Great. Can you tell me about them?
- Nestor:** Of course. First, you request access to an application resource, then the browser sends the request to a web server. After that, the second stage involves a unified pipeline which receives the request. On this state, ApplicationManager is created, as well as something called HostingEnvironment which provides information about resources. We also have a stage where response objects are created, and another one where objects are assigned to the request; the latter is processed by the HttpApplication class.
- Jessica:** Excellent! One last thing. I am sure I'll have to exemplify my explanation by providing an example in a work context.
- Nestor:** That's easy; you can tell them to imagine a company that is organized by different departments. Each department manages sensitive data and has control over items that were inherited from management. Just as in ASP.NET, there is a page directive which sets up the environment where everyone works; there is a code selection to establish the events; and finally, a page layout where you can see the final product.
- Jessica:** You're amazing! Thanks for your help!





Let's Practice!

What does the ASP.NET life cycle specify?

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

are embedded / is specified / are processed and instantiated / are tracked

1. The ASP.NET life cycle has three main aspects and these ones _____.
2. The ASP.NET life cycle _____ by the processes of the pages that produce dynamic output.
3. Applications and pages _____ in the life cycle of the ASP.NET.
4. The life cycle of the ASP.NET specifies pages that _____ and put together in a dynamic way.



Let's Practice!

Let's Talk About the Stages in the Application Life Cycle in ASP.NET.

Instructions: Circle the correct option to complete the sentences.

1. There are several stages in the application of ASP.NET life cycle:: First, there is a request access to an application resource (that / who / by) the browser sends to a web server. The second stage involves a unified pipeline (which / whose / who) receives the request. On this stage, the Application Manager and the Hosting Environment are created (which / who / where) provides information about resources. Third and fourth, the response objects are created and are assigned to the request (that / who / when) is processed by the Http Application class.



Extension Activity

Mind Mapping about the Stages of an Application Life Cycle in ASP.NET.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for more information about applications life cycle in ASP.NET.
- Identify the stages of an application life cycle and classify the information in order of importance.
- Create a mind map to explain all the stages of an applications life cycle in ASP.NET.
- Prepare an oral presentation to explain your mind map.
- Share your work with the class.

Self-Assessment

Yes No

- I identify information about applications life cycle in ASP.
- I classify the stages of an application life cycle in ASP.NET
- I create a mind map to explain all the stages.
- I prepare an oral presentation to explain my mind map.
- I share my work with the class.

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Target Words

- **assemble** (verb): to put together parts of something
e.g. That's my favorite platform to assemble applications.
- **dynamic** (adjective): that depends on power
e.g. ASP.NET life cycle specifies the processes on the ASP.NET pages that produce dynamic output.
- **cycle** (noun): a process that repeats actions after they are finished
e.g. ASP.NET life cycle specifies the processes on the ASP.NET pages that produce dynamic output.
- **embedded** (adjective): when something comes linked to another thing
e.g. There are three main aspects that are embedded.
- **directive** (noun): an instruction which is made authoritatively
e.g. There is a page directive which set ups the environment where everyone works.
- **track** (verb): to check constantly a progress
e.g. It tracks pages and put them together in a dynamic way.



Learning Points

Simple Present Passive: Affirmative.

We use sentences in **simple present passive** or **present passive voice** when we want to stand out the function of an object in a sentence rather than the subject.

When we aim to highlight the person or element that performs an action, we use active voice:

- **ASP.NET** processes pages and it produces a dynamic output to specify the life cycle.

But when it comes to highlighting the role of the object in a sentence, we need to use the passive voice:

- **Pages** are processed to specify the life cycle.
- **Pages** are processed **by** ASP.NET to specify the life cycle.

When using the passive voice the subject could be omitted, but in case you wish to mention it, you need to use the particle **by** followed by the noun.

This is the grammatical structure we need to use when building sentences in **simple present passive**:

Object + **Verb be** + **Verb in past participle** + Complement

For example:

- Life cycles **are divided** into applications and pages in ASP.NET.

Examples

- There are three main aspects that are embedded.
- On this state, ApplicationManager is created, as well as something called HostingEnvironment which provides information about resources.
- We also have a stage where response objects are created, and another one where objects are assigned to the request; the latter is processed by the HttpApplication class.



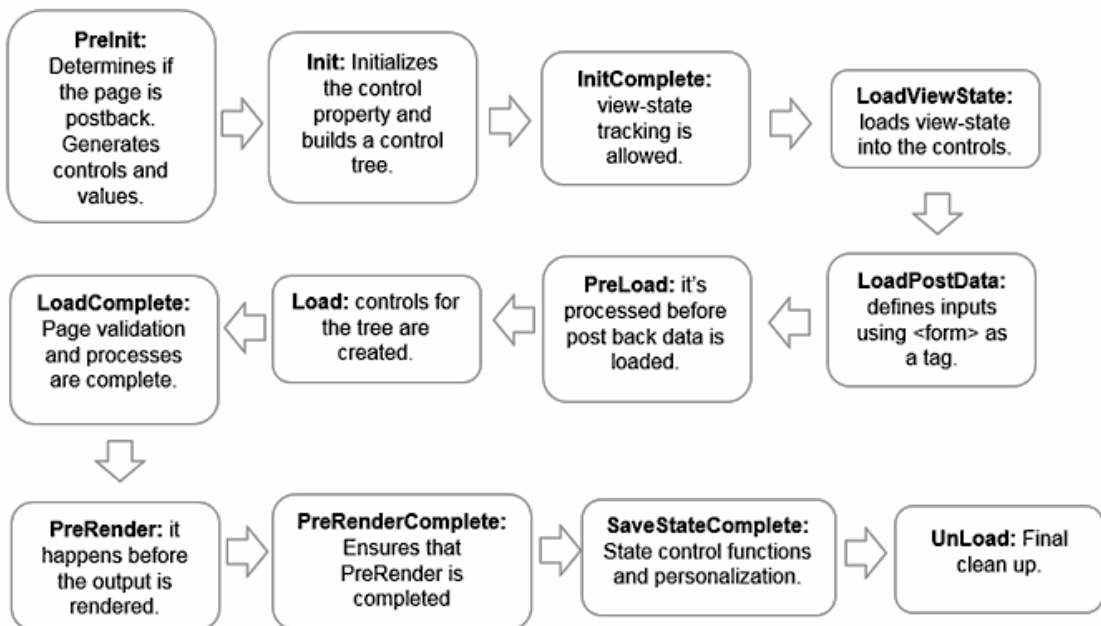
Learning Points

ASP.NET Page Life Cycle Events.

There are a series of **events** that appear at each stage of **ASP.NET page life cycle**.

During a life cycle we have some stages that present some events. These events are coded and come with an event handler which is a function or subroutine.

The following chart includes the event present in an ASP.NET page life cycle:



Example

- I know about how ASP.NET processes pages. I also know that it has some stages in the application life cycle.*